# Inorganic Cumulative Exam
## March 3, 2006

1. A pink solid has the formula $CoCl_3\ 5NH_3\ H_2O$. A solution of this salt is also pink and rapidly gives 3 mol of AgCl on titration with silver nitrate solution. When the pink solid is heated, it loses 1 mole of $H_2O$ to give a purple solid with the same ratio of $NH_3$:Cl:Co. The purple solid releases two of its chlorides rapidly; then, on dissolution and after titration with $AgNO_3$ releases one chloride slowly. Deduce the structures of the two octahedral complexes and draw and name them.

2. The rate constant for the formation of $[CoX(NH_3)_5]^{2+}$ from $Co(NH_3)_5OH_2]^{3+}$ for X = $Cl^-$, $Br^-$, $N_3^-$ and $SCN^-$ differ by no more than a factor of 2. What is the mechanism of substitution?

3. A neutral macrocyclic ligand with four donor atoms produces a red diamagnetic low-spin $d^8$ complex of Ni(II) if the anion is $ClO_4^-$. When perchlorate is replaced by thiocyanate ion, the complex turns violet and is high-spin with 2 unpaired electrons. Interpret the change in terms of structure.